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BECAMEX BUSINESS SCHOOL

BUS 450 CONCEPT NOTE

TOPIC TITLE:

End-to-End Supply Chain Analytics: Power BI Visualization and Prophet-Based Demand Forecasting Using the DataCo Smart Supply Chain Dataset

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Quarter: 4 Academic Year: 2025 - 2026

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CONCEPT NOTE

1. Project topic:

End-to-End Supply Chain Analytics: Power BI Visualization and Prophet-Based Demand Forecasting Using the DataCo Smart Supply Chain Dataset. The project will reproduce and adapt an end to end technical workflow that transforms the public DataCo dataset into a cleaned analytical dataset, SQL-based performance analysis, an interactive Power BI dashboard, and a Prophet model for 91 days (13-week) demand forecasting.

2. Related area(s):

- Supply Chain Analytics
- Power BI Data Visualization
- Demand Forecasting and Time Series Analysis
- Predictive Analytics and Data-Driven Decision Support

3. Reasons for choosing this topic:

Supply chain datasets contain large volumes of information such as transaction, products, sales, geographic, shipping and delivery that are difficult to interpret directly. Converting these records (or available datasets) into reliable indicators, dashboards and forecasts can help decision makers understand past operational performance and prepare for future demand. The project also combines descriptive and predictive analytics in a workflow.

The topic is suitable for an applied capstone because it integrates business problem definition, quantitative analysis, technical implementation, and business interpretation. It also supports my career aspirations in Supply Chain and Business Analyst field by developing practical capabilities in Python, PostgreSQL, Power BI, forecasting and giving evidence based Supply Chain knowledge.

4. Problem statement:

The DataCo Smart Supply Chain dataset contains lots of historical information about orders, quantities, sales, profits, product categories, customer information, markets, regions, shipping

modes, shipping duration and delivery outcomes. But the raw data is not immediately decision ready. Without a structured analytical process, managers may have limited visibility into revenue and demand patterns, regional performance, and delivery inefficiencies. Historical analysis alone also does not provide an estimate of near term demand, which is necessary for operational and capacity planning (Seydan & Mafakheri, 2020).

This project addresses the problem by developing an integrated pipeline for data preparation, descriptive supply chain analysis, interactive visualization, and Prophet-based demand forecasting.

The project addresses the following questions:

> **RQ1:** What sales, demand, profitability, region, shipping and late delivery patterns can be identified from the DataCo through Python, SQL analysis and Power BI visualization?
>
> **RQ2:** How accurately can Prophet forecast weekly aggregate demand over a 13-week horizon compared with a Seasonal Naive benchmark, as measured by MAE, RMSE and sMAPE?

5. Project objectives:

- Prepare a validated analytical dataset by removing duplicates, handling missing values, converting date and numeric fields, standardizing column names, excluding personally identifiable information and engineering relevant variables such as shipping delay and late-delivery status.
- Analyze operational performance using Python and PostgreSQL, and develop an interactive Power BI dashboard containing supply chain.
- Construct a weekly demand series and develop Prophet as the primary forecasting model and evaluate forecast accuracy using MAE, RMSE and sMAPE, then translate the technical results into practical supply chain recommendations.

6. What are the theoretical foundations of your Project?

This project draws on four theoretical foundations. The number one is Supply Chain Analytics, particularly how descriptive and predictive analytics turn raw operational data into decision support. Big data analytics only lifts supply chain performance when analytical resources are translated into routine decision processes (Gunasekaran et al., 2017).

The database layout provides a structured foundation for storing cleaned data and supporting reproducible analysis. Data warehouse design should align the data model with both source-data structures and analytical requirements. Therefore, this project organizes the cleaned Dataco into fact and dimension tables in PostgreSQL before connecting the database to Power BI for visualization (Hanine et al., 2021). Data visualization provides the communication layer of the descriptive analytics stage. Power BI will be used as the visualization platform because effective dashboards require careful decisions concerning information hierarchy, visual encoding, layout, color and interaction (Bach et al., 2022).

The demand forecasting itself by using past demand to cut down planning uncertainty. Forecast accuracy hinges as much on data preparation and model choice as on the algorithm itself (Seyedan & Mafakheri, 2020). For the technical core, the project uses Prophet, an additive time series model that breaks a series into trend, seasonality, and change points (Taylor & Letham, 2018). Accuracy will be checked through chronological train test splits and out-of-sample error metrics, since forecasts can only be judged fairly on data the model hasn't already seen (Hyndman & Athanasopoulos, 2021).

This project only borrows that logic, it doesn't reproduce their ensemble deep learning model or order-up-to-level optimization comes from Seyedan, Mafakheri, and Wang (2023). The actual pipeline instead follows Jahid's (n.d.) Prophet-based workflow.

7. Methodology:

7.1. Research design

The project will use a quantitative, applied and secondary data research design. It will independently reproduce and adapt an end-to-end analytics workflow using DataCo dataset. The design combines descriptive analysis for historical performance and predictive analysis for future demand.

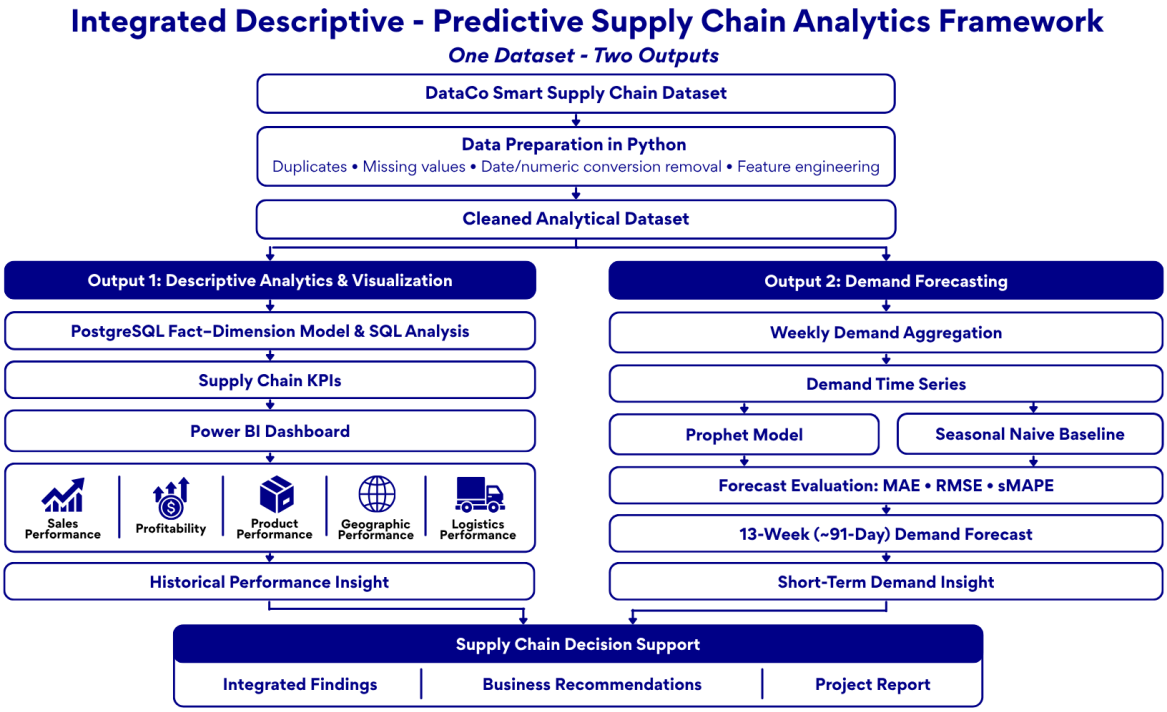


Figure 1. Integrated Descriptive - Predictive Supply Chain Analytics Framework

This figure presents the proposed end-to-end analytical workflow in which one cleaned DataCo dataset produces two complementary analytical outputs. Branch 1 uses PostgreSQL for SQL analysis and Power BI visualization, while Branch 2 aggregates weekly demand and compares the Seasonal Naive benchmark with Prophet before producing a 13-week demand forecast. Findings from both branches are integrated into practical recommendations for performance monitoring, demand forecasting and operational improvement.

7.2. Data preparation

Python and pandas will be used to inspect the dataset, remove duplicate records, assess missing values, convert order and shipping dates, validate numeric fields, standardize column names, remove unnecessary personal identifiers and derive analytical variables. The cleaned data will be exported in an efficient format and loaded into PostgreSQL for reproducible SQL analysis.

7.3. Descriptive analytics

Python and SQL will be used to calculate total orders, revenue, profit, demand, category and regional performance, shipping duration , shipping delay and late-delivery rates. Power BI will then present these measures through KPI cards, time trends, category comparisons, regional views, shipping mode analysis and interactive filters.

7.4. Demand forecasting and model evaluation

Adapting the repository's forecasting workflow, weekly demand is defined as the weekly sum of Order Item Quantity. The series will first be checked for missing periods, unusual observations, trend and seasonality before any modeling begins. Prophet serves as the primary model, with trend and seasonal components tuned to match what the data actually shows rather than left at default settings. The last 13 weeks are held out as a chronological test set roughly a 91-day horizon and rolling cross validation will be added where the data allows it. A Seasonal Naive model provides the benchmark, and both models are compared on MAE, RMSE, and sMAPE.

7.5. Analytical tools

The implementation will use Python in Google Colab with pandas and NumPy for data preparation, Prophet for forecasting and matplotlib or seaborn for diagnostic visualization, PostgreSQL will support structured business analysis and Power BI will be used for the interactive dashboard.

8. Data collection:

The project uses secondary quantitative data from DataCo Smart Supply Chain for Big Data Analysis, version 5, published on Mendeley Data under a CC BY 4.0 license (Constante et al., 2019). The main file, DataCoSupplyChainDataset.csv holds approximately 180,500 order-item records covering 65,752 unique orders from 2015 to early 2018. Key fields used span order, shipping dates, Order ID, Order Item Quantity, sales and profit figures, product/category details, market and region, shipping mode and delivery related variables such as scheduled versus actual shipping days, delivery status and late delivery risk. The dataset's separate click stream file falls outside this project's scope, since the repository-based forecasting workflow being reproduced doesn't call for it. Identifying fields including

customer names, email addresses, passwords, street addresses will be dropped before any analysis as they play no role in the forecast and aren't needed for the study's purpose.

9. Other relevant information:

Expected outputs

1. **Output 1:**
    - A cleaned and documented DataCo analytical dataset.
    - A reproducible PostgreSQL database with a documented fact and dimension model and SQL business-query file.
    - An interactive Power BI dashboard covering revenue, orders, demand, profitability, regional performance, shipping modes, and late delivery indicators
2. **Output 2:**
    - A Prophet model producing a 13-week aggregate demand forecast with uncertainty intervals, diagnostic plots, and out of sample accuracy results.
    - A comparison between Prophet and the Seasonal Naive benchmark using MAE, RMSE, and sMAPE.

10. References

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